

Exact Primal-Dual Certificates for a Two-Bidder, Two-Item Auction

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Abstract

Multidimensional incentive constraints couple auction outcomes across reports, so optimizing the two bidders' menus separately need not produce an optimal feasible auction. We study two additive bidders and two items with independent uniform values and combine exact global upper certificates with residual- capacity screening. An explicit randomized, pointwise DSIC and ex-post IR mechanism has revenue $0.875819854148422455346\dots$, while the unrestricted upper bound is $0.8857583025186545848846\dots$. The certified gap is below 0.01, and the mechanism earns at least 98.8779% of unrestricted optimal revenue. The new analytic components are finite capacity-price measures supporting full randomized conditional optimization at a diagonal capacity hole and a lottery junction. Their exact slack identities retain singular report-line constraints and unnormalized information rents. Binding capacity conditions then guide complete joint reallocations with exact algebraic and logarithmic revenue increments. A profitable two-sided change demonstrates why conditional optimality alone does not supply a common auction certificate. The global upper proof uses a rational stream correction and independently replayed tensor-Bernstein enclosure. Our lower bound remains below the externally reported GemNet revenue of approximately 0.876; the unrestricted optimal auction and a matching common certificate remain open.

1 Introduction

Optimal auction design is completely understood for a single item under the classical regularity conditions of Myerson [6]. With several items, bidder types become multidimensional and dominant-strategy incentive compatibility imposes global cross-type restrictions. Even the canonical case of two additive bidders, two items, and independent uniform values has resisted an exact characterization [4, 8].

Computational work has produced strong feasible mechanisms and rigorous upper bounds. Affine-maximizer searches provide transparent deterministic benchmarks [8], while GemNet constructs fully strategyproof menu mechanisms after a compatibility transformation [9]. Jiang, Parkes, and Wang report approximately 0.876 revenue for that GemNet benchmark on the present instance. On the dual side, the same paper develops flow-conserving DSIC kernels and a continuous lifting theorem; its strict 50×50 uniform continuous certificate is 0.8919 [4]. These are external benchmarks. The GemNet value is a reported computational revenue, whereas 0.8919 is a rigorous upper certificate. We improve the upper benchmark and provide explicit mechanisms with exact revenue accounting. The latest randomized construction narrows our certified interval below 0.01 without claiming to improve GemNet's reported revenue.

We pursue a target-specific continuous witness rather than another grid lift. Pointwise DSIC represents each bidder's truthful utility as a convex potential [7]. A radial integration identity produces a base vector field with the required distributional source. We then add a rational

polynomial curl generated by a stream function. The correction has zero divergence and zero normal boundary component, so it changes neither the conservation law nor the revenue pairing. Ex-post feasibility bounds the paired allocation pointwise by the positive maximum of the two bidders' field components.

The remaining problem is a four-dimensional continuous polynomial-envelope integral. We certify it using tensor Bernstein control coefficients, exact rational construction, directed fixed-point rounding, and a certificate-aware nonuniform dyadic traversal concentrated near unresolved winner-switching boxes. The certificate proves the upper bound $0.8857583025\dots$. A separate implementation reconstructs the polynomial charts and repeats the complete depth-21 base traversal plus selective depth-22 and depth-23 refinement without importing the formal verifier. On the primal side, the earlier deterministic layers use deletion containment to preserve feasibility. The new construction allows ownership transfers. We fix one bidder's complete mechanism, solve selected opposing conditional screening problems under the actual residual, and derive measure-valued supports from their binding incentive and capacity constraints. The supports act on whole convex utility traces rather than on virtual revenue at isolated types. Their failure to support the other bidder guides a complete joint change whose induced payment and information-rent effects can be integrated exactly.

Contributions. The paper develops four connected components:

1. an explicit rational stream weak-duality witness for the full randomized pointwise-DSIC class, with a separate exact Bernstein enclosure and a non-importing implementation replay;
2. full randomized conditional screening certificates for a diagonal capacity hole and a lottery junction, using nonnegative finite measures with volume and line components and exact nonnegative slack identities;
3. complete feasible joint reallocations guided by those supports, including a coupled singleton/bundle response and a profitable lottery-price cut compensated by entry fees on complete opposing menus; and
4. an explicit randomized mechanism with exact revenue specification, a certified bracket of width below 0.01, and a 98.8779% revenue guarantee.

The conditional theorems impose no finite-menu restriction on competitors. They identify optimizers for stated residuals, not the optimizer of the full auction. The distinction is witnessed by the strictly profitable joint change itself.

2 Relation to prior dual formulations

Rochet's cyclic-monotonicity characterization turns DSIC into convexity of truthful utility and identifies allocation vectors with subgradients [7]. Continuous duality for the one-bidder multi-item problem has been developed through transport and partial-derivative formulations [2, 3]. These results motivate integration-by-parts certificates but do not directly enforce item supply across several bidders.

Multi-bidder dual formulations also predate the present work. Finite-type flow-induced virtual values arise in unified Bayesian mechanism-design duality [1]; Zuo considers both Bayesian and dominant-strategy formulations [10]; and Beckmann-style transport gives a continuous strong-duality formulation for multi-item, multi-bidder BIC auctions in interim space [5]. The latter problem uses weaker incentive and individual-rationality requirements than the pointwise DSIC and ex-post IR requirements studied here.

Most directly, Jiang, Parkes, and Wang construct nonnegative, flow-conserving DSIC deviation kernels, derive continuous weak duality, and lift coarse uniform-grid certificates to continuous type spaces [4]. Our proof shares the same high-level weak-duality architecture: conservation eliminates payments, the dual object induces virtual coefficients, and ex-post feasibility bounds virtual surplus itemwise. The representation and certification are different. We use a signed vector field with a distributional radial source plus a divergence-free tangent stream correction, and prove its validity directly against Lipschitz convex utilities. We do not assert that every such stream field is generated by a nonnegative deviation kernel, or conversely. The contribution is an explicit rational witness for this canonical DSIC instance, an exact Bernstein certificate for its continuous objective, and explicit conditional capacity supports that guide feasible two-bidder reallocations. The convex-mixture concavity of residual value is standard infrastructure; the new theorem content is the displayed residual-specific measures, their full-class slack identities, and the resulting complete primal variations. We make no first-in-literature priority claim for the general use of convex potentials, transport, or complementary slackness.

3 Setting and proved bounds

Write a report profile as

$$(x, y, z, w) = (v_{11}, v_{12}, v_{21}, v_{22}) \in [0, 1]^4.$$

The two bidders are risk-neutral and additive; the two items are indivisible. The four coordinates are independent and uniform. A direct mechanism specifies jointly measurable expected allocation probabilities $x_{ij} : [0, 1]^4 \rightarrow [0, 1]$ and expected payments $p_i : [0, 1]^4 \rightarrow \mathbb{R}$, with $p_i \in L^1([0, 1]^4)$. At every report profile, itemwise feasibility requires $x_{1j} + x_{2j} \leq 1$. These marginals can be realized by assigning each item to bidder 1, bidder 2, or the seller with the corresponding probabilities. DSIC and ex-post IR are required at every type and opponent report, with utilities averaged over the mechanism's internal randomization. Let OPT be the supremum of expected revenue over this class. Our constructed mechanisms are Borel. The upper argument also covers joint Lebesgue measurability with integrable payments: on every opponent fiber, DSIC supplies a convex Lipschitz utility and the selected allocation equals its gradient off an own-type null set; Fubini supplies the integrated inequality. No pointwise Borel-representative repair is asserted. For the coordinate-line supports below, tangential selections are monotone and their line integrals are well-defined. Residual functions are kept pointwise, not identified modulo area-null sets.

Theorem 1 (Audited sub-0.01 bracket). *An explicit admissible randomized mechanism has exact revenue R_J specified in (7), with*

$$0.8758198541484224553460 < R_J < 0.8758198541484224553461.$$

For the unrestricted class,

$$R_J \leq \text{OPT} \leq U_{\text{stream}} = \frac{3715139591287203}{4194304000000000} = 0.8857583025186545848846435546875,$$

and

$$0 < U_{\text{stream}} - R_J < 0.0099384483702321295386 < \frac{1}{100}, \quad \frac{R_J}{\text{OPT}} \geq \frac{R_J}{U_{\text{stream}}} > 0.988779.$$

Thus the mechanism earns at least 98.8779% of unrestricted optimal revenue. The upper follows from the retained rational stream witness and finite exact enclosure. The lower is a complete jointly feasible mechanism, not a grid solution. These bounds do not identify OPT or decide its attainment.

The lower bound is proved in Sections 4–6; the upper bound and a simpler radial benchmark are proved in Section 7. The stream construction supplies a valid upper certificate without assuming completeness of the chosen witness family. As an external comparison, Jiang, Parkes, and Wang report a strict continuous upper certificate 0.8919 and fully strategyproof GemNet revenue approximately 0.876 for this instance [4]. Our stream bound strictly improves their printed upper decimal by the exact rational difference

$$\frac{8919}{10000} - U_{\text{stream}} = \frac{25760146312797}{4194304000000000} > 0.$$

This comparison is to their reported decimal, not a reconstruction of their underlying dual array. The 0.876 GemNet revenue is likewise a reported computational benchmark, not an exact rational certificate replayed here.

4 An exact base affine maximizer

Set

$$a = \frac{159}{250}, \quad b = \frac{91}{100}, \quad s = \frac{1137}{1000}.$$

For each feasible outcome A , let c_A and the affine score be as follows. The row number is also the tie-breaking priority.

id	allocation	score	c_A
0	seller keeps both	0	0
1	$1 \rightarrow 1$	$x - a$	a
2	$2 \rightarrow 1$	$y - a$	a
3	$\{1, 2\} \rightarrow 1$	$x + y - b$	b
4	$1 \rightarrow 2$	$z - a$	a
5	$2 \rightarrow 2$	$w - a$	a
6	$\{1, 2\} \rightarrow 2$	$z + w - b$	b
7	$1 \rightarrow 1, 2 \rightarrow 2$	$x + w - s$	s
8	$2 \rightarrow 1, 1 \rightarrow 2$	$y + z - s$	s

The base outcome $A^0(v)$ is the smallest-id maximizer of the nine scores. This defines the outcome on every equality surface.

Let

$$H_1(z, w) = \max\{0, z - a, w - a, z + w - b\}, \quad H_2(x, y) = \max\{0, x - a, y - a, x + y - b\}.$$

If $V_i(A)$ is bidder i 's reported value for the items assigned in A , the base payments are

$$p_1^0 = H_1 - V_2(A^0) + c_{A^0}, \quad p_2^0 = H_2 - V_1(A^0) + c_{A^0}.$$

Lemma 2. *The base mechanism is feasible, pointwise DSIC, and pointwise ex-post IR.*

Proof. Every listed outcome is feasible. Fix bidder 2's report. Apart from the constant H_1 , bidder 1's utility from an induced outcome A , evaluated at the true type, is

$$V_1(A) + V_2(A) - c_A - H_1.$$

Truthful reporting selects a maximizer of the expression preceding H_1 ; no report can give greater utility. The same argument applies to bidder 2. It remains valid on ties because the selected row is still a maximizer. The outcomes excluding bidder 1 attain H_1 , so the winning score is at least H_1 , and bidder 1's truthful utility is nonnegative. The symmetric argument proves bidder 2's IR. $\square$

For a bidder facing opponent report $q = (q_1, q_2)$, define

$$H(q) = \max\{0, q_1 - a, q_2 - a, q_1 + q_2 - b\}.$$

The base taxation menu has prices

$$A_0(q) = H(q) + \min\{a, s - q_2\}, \quad B_0(q) = H(q) + \min\{a, s - q_1\}, \quad C_0(q) = H(q) + b \quad (1)$$

for item 1, item 2, and the bundle, respectively; opt-out costs zero. These formulas follow directly by minimizing the outcome cost net of the opponent's value among outcomes assigning the indicated bundle.

5 The deterministic entry-surcharge predecessor

Opponent-conditioned taxation menus ensure each bidder's DSIC, but their separate choices need not be simultaneously feasible. The following containment principle supplies that link.

Lemma 3 (Deletion containment). *Let a deterministic predecessor mechanism have feasible allocations $S_i^0(v)$. For each bidder and opponent report, fix a finite taxation menu containing opt-out at price zero. Suppose its declared tie rule selects a utility-maximizing bundle $S_i(v) \subseteq S_i^0(v)$ at every report profile. Then the resulting mechanism is pointwise DSIC, ex-post IR, and ex-post feasible.*

Proof. Fixing the opponent report fixes all available bundles and prices. A truthful utility maximizer cannot gain by reporting another type, and opt-out guarantees nonnegative utility. Containment preserves the predecessor's disjoint item allocations. The assumptions concern the selected bundles, including ties, so all three conclusions hold pointwise. $\square$

For an opponent report $q = (q_1, q_2)$, set

$$t = \max\{q_1, q_2\}, \quad \rho = \min\{q_1, q_2\}.$$

The active mechanism is the finite rational rule frozen in

`certificate/refined_item_containment_bundle_pivot_lower_bound/manifest.json`.

The complete evaluable tables and endpoint conventions are specified by this manifest and its hash-bound predecessor manifests. The active manifest contains the eight item-containment rows. The twenty Z/S rows and 41 bundle-pivot rows are specified, respectively, in

`certificate/piecewise_surcharge_twenty_band_lower_bound/manifest.json`
`certificate/piecewise_surcharge_bundle_pivot_lower_bound/manifest.json`.

We summarize the three-layer construction here.

First, twenty rational Z/S half-bands add a common fee $F(q)$ to all three nonempty prices in (1). If none matches, set

$$u = t + \rho, \quad c = b - a = 137/500, \quad d = s - a = 501/1000,$$

and first test the bundle-pivot region

$$91/100 \leq u \leq 1, \quad c \leq \rho \leq u - d.$$

Only inside this region define $r = (\rho - c)/(u - c - d)$, whose denominator satisfies $u - c - d \geq 27/200 > 0$, and test the 41 positive cells in (u, r) . Outside the region no bundle-pivot fee is applied and r need not be evaluated. Those cells also add one common fee to all nonempty prices. The Z/S rows have priority on shared boundaries. Their interiors are disjoint from the

bundle cells: Z support has $u \leq b$ and S support has $\rho \leq c$, whereas a bundle-cell interior has $u > b$ and $\rho > c$.

Finally, eight rows inside the two S5 half-bands add $\delta(q)$ to the singleton corresponding to the opponent's low coordinate and to the grand bundle, leaving the other singleton unchanged. This coordinate is unique: on every item row, $t > 139/200 > 137/500 \geq \rho$. Each row is left-open and right-closed in t ; at $t = 139/200$, $\delta = 0$. If maximum utility is zero, choose opt-out. At a positive tie, retain the predecessor-selected outcome whenever it remains maximizing; otherwise choose the maximizing itemwise subset with smallest id under the fixed predecessor order.

Proposition 4. *The refined surcharge mechanism is deterministic, measurable, ex-post feasible, pointwise DSIC, and pointwise ex-post IR on the entire cube, including all menu, band, cell, and item-surcharge boundaries.*

Proof. Every common fee preserves the ranking among nonempty bundles, so its selected bundle is the base bundle or opt-out. On an item row, orient the predecessor prices as

$$(A, B, C) = (d + F, t + F, t + c + F).$$

The row changes them to $(A + \delta, B, C + \delta)$, and its exact verifier checks

$$0 < \delta < A + B - C = d - c + F.$$

If the changed singleton was selected, its utility lead over the unchanged singleton is at least $A + B - C$, and hence remains positive after the change; its ranking against the bundle is unchanged. The unchanged singleton remains selected when it was selected, a bundle buyer can move only to a subset, and an opt-out cannot enter because no nonempty price falls. Thus every final bundle is an itemwise subset of its predecessor bundle, which is itself a subset of the feasible base allocation.

For each fixed opponent report the declared rule selects a utility maximizer from a taxation menu independent of the bidder's report, with opt-out at price zero. Lemma 3 therefore proves DSIC, ex-post IR, and simultaneous feasibility at each layer. The finite rational semialgebraic tables and literal endpoint priority are Borel measurable. Numerical searches selected rational rows and fees, but no discovery file is a verifier input. $\square$

6 Exact revenue and residual screening

Lemma 5 (Revenue of a four-option menu). *Suppose*

$$0 \leq A, B \leq C \leq 1, \quad C \leq A + B.$$

Let an additive type be uniform on $[0, 1]^2$. The expected payment from the four-option menu is

$$\begin{aligned} r(A, B, C) &= A(1 - A)(C - A) + B(1 - B)(C - B) \\ &\quad + C \left[(1 - C + B)(1 - C + A) - \frac{(A + B - C)^2}{2} \right]. \end{aligned}$$

Proof. The strict item-1 region has area $(1 - A)(C - A)$, and the strict item-2 region has area $(1 - B)(C - B)$. The bundle region starts with the rectangle $[C - B, 1] \times [C - A, 1]$ and removes a right triangle of side $A + B - C$. Multiplying the three areas by their prices proves the formula. Ties have area zero for this integral; their pointwise allocation was already fixed in Proposition 4. $\square$

Suppose both (A, B, C) and $(A + e, B + e, C + e)$ satisfy Lemma 5. For the nonnegative common surcharges used here, the original price conditions together with $C + e \leq 1$ ensure this. Direct expansion then gives the actual revenue change

$$r(A + e, B + e, C + e) - r(A, B, C) = eG - \frac{3C}{2}e^2 - \frac{1}{2}e^3, \quad (2)$$

where

$$G = (C - A)(1 - 2A) + (C - B)(1 - 2B) + (1 - C + A)(1 - C + B) - \frac{(A + B - C)^2}{2} - C(A + B - C).$$

For a Z-band orientation the prices and cross-section length are

$$(A, B, C) = (s - t, a, b), \quad b - t,$$

and for an S-band orientation they are

$$(A, B, C) = (t, s - a, t + b - a), \quad b - a.$$

Exact integration of the twenty frozen half-bands gives

$$G_Z = \frac{113291698497241117}{10000000000000000000}, \quad G_S = \frac{2400553481931223}{31250000000000000000},$$

and therefore the common-fee gain

$$G_Z + G_S = \frac{190109409919040253}{100000000000000000000}.$$

On a bundle-pivot cell the prices are

$$(A, B, C) = (u - c, \rho + 227/1000, u).$$

The normalized map has Jacobian $u - 31/40$; including both orientations and both bidders gives symmetry factor four. Direct bivariate integration of the 41 positive cells gives

$$G_B = \frac{70946101529751}{102400000000000000000}.$$

The eight item-containment rows give

$$G_I = \frac{3963982653557301}{800000000000000000000}.$$

The base mechanism's exact expected revenue is

$$R_0 = \frac{26232089810531183}{30000000000000000000}.$$

The exact verifier in `certificate/ama_lower_bound` reconstructs the $9 \cdot 4 \cdot 4 = 144$ affine/pivot cells using rational arithmetic, finds 39 full-dimensional cells, triangulates them into 16,336 rational four-simplices, checks total volume one, and integrates the affine payments.

The added gains are integrated layer by layer. The Z/S and bundle-pivot supports have disjoint interiors. Item-containment rows lie within the S5 support, but G_I is the change relative to the already surcharged predecessor menu, not relative to the base. Thus the revenue sum telescopes without double counting.

The active primary checker binds every predecessor by SHA-256, directly reconstructs the eight item-containment price regimes, checks their deletion slack and coverage, and integrates their exact symbolic polynomials. Its non-importing replay intersects the linear utility-comparison half-planes with $[0, 1]^2$. At rational prices, the resulting demand polygons have rational vertices, so their areas and conditional revenues are computed exactly. On each item row, the prices are affine in t and the price-regime inequalities hold throughout the interval by exact endpoint checks. Lemma 5 therefore makes each conditional revenue a polynomial of degree at most three in t . Boole’s rule, which integrates polynomials of degree at most five exactly, then gives the row integral in rational arithmetic. The common-fee rows remain within that degree range after multiplication by their affine cross-section lengths. On bundle cells, the prices are affine in each of u, r ; including the Jacobian gives degree at most four in u and three in r , so tensor Boole quadrature is exact. Finite differences at the sampled rational points are consistency checks, not proofs of a polynomial degree bound on the whole interval. The hash-bound sealed predecessors separately recompute the 20 common-fee rows and 41 positive bundle-pivot cells; the bundle predecessor has its own direct bivariate integration and independent scalar tensor-quadrature replay. These replays independently recompute the added gains, but share the base value R_0 certified by the single polytope verifier. Table 1 summarizes the dependencies. SHA-256 bindings establish the identity of these shared inputs.

Component	Revenue term	Exact integration and replay
Affine base	R_0	One rational polytope verifier
Twenty common-fee rows	$G_Z + G_S$	Polynomial integral; scalar replay
41 bundle-pivot cells	G_B	Bivariate integral; tensor replay
Eight item rows	G_I	Polynomial integral; polygon replay

Table 1: Revenue decomposition. Each added term is measured against its predecessor; all added-gain replays share the same certified base revenue.

Therefore

$$R_0 + G_Z + G_S + G_B + G_I = L_{\text{det}} = \frac{83962078694672281756033}{9600000000000000000000000000},$$

certifying the retained deterministic predecessor.

The bundle-pivot, twenty-band, ten-band, and former two-rectangle certificates are retained as reproducible predecessors in their separate directories. The construction below is the active lower mechanism.

6.1 Residual capacity as a screening constraint

Deletion containment gives a convenient sufficient condition for feasibility, but excludes profitable changes in ownership. We instead fix a complete mechanism for bidder 1 and optimize bidder 2 against the residual allocation. For a jointly measurable capacity function $r : [0, 1]^4 \rightarrow [0, 1]^2$, let

$$\mathcal{V}(r) = \sup\{\text{Rev}_2(M_2) : M_2 \text{ is pointwise DSIC/IR and } x_2 \leq r\}.$$

The mechanisms in this definition remain jointly measurable, with integrable payments. Grouping feasible pairs by their first component gives exactly

$$\text{OPT} = \sup_{M_1} \{\text{Rev}_1(M_1) + \mathcal{V}(1 - x_1)\}.$$

This identity does not assert existence of a best response or invoke a measurable selection of pointwise optimizers.

Proposition 6 (Residual value and supporting measures). *The functional $\mathcal{V}$ is nondecreasing and concave. On a fixed opponent fiber, suppose a finite nonnegative capacity-price measure π satisfies $R(u, a) \leq \langle \pi, a \rangle$ for every admissible conditional mechanism, and a feasible candidate at r^* attains $R^* = \langle \pi, r^* \rangle$. Then that candidate solves the full randomized conditional problem and, for every residual for which the pairing is defined,*

$$V(r) \leq V(r^*) + \langle \pi, r - r^* \rangle.$$

Proof. Monotonicity is immediate. Mix any two admissible mechanisms using one exogenous coin with constant probability θ . All pointwise incentive, IR and capacity inequalities are preserved, as are measurability and integrability. Apply this to arbitrarily near-optimal competitors and pass to the supremum. For the second assertion, $R \leq \langle \pi, a \rangle \leq \langle \pi, r \rangle$, and equality at r^* gives the claim. No own-report-dependent mixture is used. $\square$

Concavity is the standard convex-mixture observation. The substantive step below is an explicit support for residuals with binding report-line constraints. These supports price information rents as well as item supply. They are conditional certificates; a single measure supporting both bidders simultaneously is a separate, unresolved requirement.

6.2 A diagonal-hole capacity certificate

Use aligned own coordinates (x, y) , where item 1 is scarce. Put $A = 2/3$, $k = C - B < 1/2$, $0 \leq k$, and

$$B = C - k \leq A, \quad C_0 = 5/6 + 3k^2/4, \quad C \geq C_0.$$

The candidate menu has prices $(0, A, B, C)$ and utility $u^* = \max(0, x - A, y - B, x + y - C)$. Assume its complete allocation is feasible against a Borel residual r^* and that (the first two conditions are needed only when $m = 2(C - C_0) > 0$)

$$\begin{aligned} r_1^*(x, 0) &= 0 \quad (0 < x < 1/2), \\ r_2^*(1/2, y) &= 0 \quad (0 < y < C - 1/2), \\ r_1^*(x, 1) &= 0 \quad (0 < x < k) \quad \text{if } B < A. \end{aligned} \tag{3}$$

Define $\ell(x) = B$ below k and $\ell(x) = C - x$ on (k, A) ; let $D_0 = \{x < A, y < \ell(x)\}$. Set $f = 3\ell - 2$ below A and $f = 1$ above A . Direct integration gives

$$m = \int_0^1 f = 2(C - C_0) \geq 0, \quad 3|D_0| = m + 1.$$

The function $F_m(x) = \int_x^1 f(s) ds - m\mathbf{1}_{x < 1/2}$ is nonnegative. Indeed, below the anchor it equals $-\int_0^x f$, while above the anchor its tail is nonnegative because $f \leq 0$ below A and $f = 1$ above A .

Theorem 7 (Anchored full screening support). *Under these hypotheses the candidate solves the complete randomized conditional problem. A supporting capacity measure is*

$$\begin{aligned} \pi_1 &= 3(x - A)_+ dx dy + F_m(x) dx \delta_1(dy) + m\mathbf{1}_{0 < x < 1/2} dx \delta_0(dy), \\ \pi_2 &= 3(y - \ell(x))_+ \mathbf{1}_{x < A} dx dy + m\delta_{1/2}(dx) dy. \end{aligned}$$

For every randomized DSIC/IR competitor with selected allocation $a \leq r$,

$$\langle \pi, r \rangle - R = \langle \pi, r - a \rangle + 3 \int_{D_0} u - mu(0, 0) \geq 0. \tag{4}$$

Proof. Uniform revenue is $R = \int u(1, y) dy + \int u(x, 1) dx - 3 \int u$. Horizontal envelopes on $x > A$ and vertical envelopes above $\ell(x)$ give the displayed volume prices and the residual top pairing $\int f(x)u(x, 1) dx$. Anchoring it at $1/2$ gives

$$\int f(x)u(x, 1) dx = mu(1/2, 1) + \int F_m(x)a_1(x, 1) dx.$$

Expand $u(1/2, 1)$ along the bottom edge and then the vertical line at $1/2$. This produces the remaining two prices and (4). Monotonicity and IR imply $u \geq u(0, 0) \geq 0$, so the non-capacity slack is at least $u(0, 0)$ by $3|D_0| = m + 1$.

At the candidate this slack vanishes. Every positive volume price charges a unit marginal. The top price below k is zero if $B = A$ and is saturated by the last condition in (3) otherwise. The other two conditions saturate the bottom and vertical charges. All line endpoints have zero price mass, but retain their complete allocation rules. $\square$

This proof uses the actual tangential allocation on each priced line. Pointwise DSIC makes that component monotone and equal to the convex trace's derivative almost everywhere on the line. It therefore covers arbitrary selected subgradients and allocations with an infinite range. It neither normalizes $u(0, 0)$ to zero nor restricts competitors to a finite menu. At $B = A$, $C = b_0 = (4 - \sqrt{2})/3$, the mass m is zero and no occupied trace is required. This recovers the full-capacity single-buyer support.

6.3 A lottery-junction capacity certificate

Set $c = 137/500$, $q = 113/500$, $T = 613/750$, and $A < t < T$. Write

$$\alpha = 3t - 2, \quad \delta = q + 3q^2/4 - 3qt/2 + \alpha^2/16, \quad \beta = \frac{\alpha}{\alpha + 2\delta},$$

$$B = 1/2 + \delta, \quad C = t + c + \delta, \quad k = t - q, \quad \ell_* = \delta + \alpha/2, \quad Y = c + \ell_*, \quad j = 1 - t/2.$$

Here $0 < k < j < A < t < 1$, $c < Y < B < A$ and $0 < \beta < 1$. Offer, in order,

$$(0, 0; 0), (0, 1; B), (1, 0; t), (1, 1; C), (1, \beta; t + \beta c),$$

choosing the first utility maximizer. Assume feasibility at r^* and

$$r_1^*(x, 1) = 0 \ (x < k), \quad r_1^*(x, c) = 0 \ (A < x < t). \quad (5)$$

Define $z(y)$ by t below c , $t - \beta(y - c)$ on (c, Y) , and A above Y . Define $e(x)$ by B below k , $C - x$ on (k, j) , and Y on (j, A) . Horizontal integration on $G = \{x > z(y)\}$ and vertical integration on $H = \{x < A, y > e(x)\}$ partition the positive-utility region up to volume-zero faces. Put $D_0 = [0, 1]^2 \setminus (G \cup H)$; the candidate has zero utility *almost everywhere* on D_0 . This statement about a volume integral does not discard pointwise capacity or line prices.

Let $f = 3e - 2$ below A and $f = 1$ above A . The displayed formula for δ implies $\int f = 0$, and $F(x) = \int_x^1 f \geq 0$. Set $\lambda = \alpha(c + \ell_*/4)$ and let $w(x)$ equal zero below A , $3t(c + \ell_*/4)$ on (A, t) , and λ above t .

Theorem 8 (Full randomized lottery support). *Under these hypotheses the five-option candidate is optimal over all randomized conditional mechanisms. A supporting measure is*

$$\pi_1 = 3(x - z(y))_+ dx dy + F(x) dx \delta_1(dy) + w(x) dx \delta_c(dy),$$

$$\pi_2 = 3(y - e(x))_+ \mathbf{1}_{x < A} dx dy.$$

For every competitor with $a \leq r$,

$$\langle \pi, r \rangle - R = \langle \pi, r - a \rangle + T_g + \lambda M_a + S_u \geq 0. \quad (6)$$

Proof. Let $g(y) = u(1, y)$ and let ν be its nonnegative second-derivative measure. The horizontal and vertical envelopes leave top coefficient f and right coefficient $\sigma(y)$, equal to α below c , $\alpha - 3\beta(y - c)$ on (c, Y) and zero above Y . The top coefficient integrates to the stated top-line price. Convexity on the whole lottery interval gives

$$\begin{aligned} T_g &= \lambda g(c) - \int \sigma(y) g(y) dy \\ &= \alpha \int_0^c [g(c) - g(y)] dy + \frac{\beta}{2} \int_0^{\ell_*} s(\ell_* - s)^2 d\nu(c + s) \geq 0. \end{aligned}$$

This follows from the exact hinge coefficient $\int_s^{\ell_*} (\alpha - 3\beta z)(z - s) dz = -\beta s(\ell_* - s)^2/2$ and $\beta \ell_* = \alpha/2$. Next, monotonicity of $a_1(\cdot, c)$ gives

$$M_a = \frac{t}{t - A} \int_A^t a_1(x, c) dx - \int_0^t a_1(x, c) dx \geq 0.$$

Expand $g(c)$ along this line to obtain its price w . Finally, $S_u = 3 \int_{D_0} u - \lambda u(0, c) \geq 0$ because D_0 contains $[0, k] \times [c, B]$ up to volume-zero faces and

$$3k(B - c) - \lambda \geq q(1 - 3q/2) > 0.$$

These identities yield (6) without removing origin utility. At the candidate the trace is constant below c , affine on (c, Y) , and all three non-capacity slacks vanish. Feasibility and (5) saturate the prices. The safe-item price is zero on the genuine lottery cell, consistent with its fractional allocation. $\square$

The hinge term excludes nonlinear convex trace changes and additional lotteries, including scarce allocations below one. Thus the finite menu is an equality case of a bound over the unrestricted conditional class, not a without-loss restriction imposed on that class.

6.4 Complete joint reallocations

The archived reference is the explicit V4.5 mechanism, defined by its finite menu rules, rational tables and algebraic tests in the source-bound certificate `joint_residual_screening_lower_bound`. Its complete predecessor proofs and exact revenue integrals are included in that package. The following modifications are applied in order. All new menu selections have the stated fixed priority; unmodified rows retain their complete predecessor maximizer selection, which need not be independent of the own report. Selecting any utility-maximizing option from a fixed opponent menu is sufficient for pointwise DSIC.

First extend both lottery menus to $E^+ = \{A < \max(v) < T, \min(v) < c\}$. Retain the predecessor on the $\min(v) = c$ face. The retained competing bidder cannot take its low item there, and the two occupied traces in (5) remain in place. Theorem 8 therefore solves both corresponding inner problems at this reference residual.

Second, bidder 2 receives nothing at every opponent report whenever its own type lies in the closed set

$$F = \{\max(v) \leq 1/2, v_1 + v_2 \leq b_0\}, \quad b_0 = (4 - \sqrt{2})/3.$$

On opponents in F , replace bidder 1's constant menu (a, a, b) by the full single-buyer optimal menu (A, A, b_0) . Here $a = 159/250$, $b = 91/100$. Capacity is entirely free on these fibers; zero-utility ties select empty.

Third, on $G = \{\max(v) \leq 1/2, b_0 < v_1 + v_2 < b\}$, bidder 1 offers bundle price $v_1 + v_2$, retaining singleton prices (a, a) . Bidder 2's complete response changes on $Q = \{\max(w) \leq A, \sum w \leq b\}$. For $\max(w) \leq 1/2$ its prices are $(A, A, \max(b_0, \sum w))$. In a constrained row write $t = \max(w) > 1/2$, $k = t - q$, $C_0 = 5/6 + 3k^2/4$, and $z = \sum w$. If $z > C_0$, the aligned prices are

$(A, z - k, z)$; otherwise retain its earlier Q menu. Moving the safe and bundle prices together preserves $C - B = k$. Relative to the earlier feasible Q menu this only discards allocations. A new bidder-1 bundle requires $\sum w > \sum v$, and the opposing bundle threshold prevents double allocation. At $(w, v) = ((.45, .45), (.44, .44))$, ownership transfers strictly from bidder 2 to bidder 1, who pays .88.

On the reopened Q rows, F/G bundle purchases supply the bottom and vertical conditions in (3); the top hole is inherited from the previous capacity rule. Theorem 7 recertifies the whole response. This includes all Q after combining unchanged and changed rows. It also shows why increasing only the opposing bundle price was insufficient. The complete constrained conditional revenue change is exactly $-(z - C_0)^2$ at the coupled response.

Finally, the conditional lottery certificate identifies a consumed junction that admits a profitable joint capacity release. Use one physical orientation and set

$$\varepsilon = 9/10000, \quad W = [.7, .71] \times [0, 1/5], \quad S = [.7 - 4\varepsilon, .71] \times [c - 2\varepsilon, c + 4\varepsilon].$$

The rectangles are closed. On bidder-2 menus indexed by $w \in W$, lower only the lottery payment by ε . On bidder-1 menus indexed by $v \in S$, add ε to every nonempty option. Choose empty whenever the old maximal utility is at most ε ; otherwise retain its old maximizer and charge the additional fee. All other rows are unchanged.

Proposition 9 (Pointwise joint release). *The composed mechanism is pointwise DSIC, ex-post IR and jointly feasible. Its allocations are jointly measurable and its payments are integrable. The final release has strictly positive exact revenue increment J given below.*

Proof. Each change specifies a complete opponent-dependent menu before maximizing; the empty option and full predecessor rules handle every exceptional report. For the final change only a newly chosen lottery can introduce a capacity conflict, since the fee contracts bidder 1's allocation. Writing $v = (x, y)$, its comparisons with singleton, bundle and empty force

$$c - \varepsilon/\beta \leq y \leq Y + \varepsilon/(1 - \beta), \quad x + \beta(y - c) - t + \varepsilon > 0.$$

Throughout W, $3/5 < \beta < 3/4$ and $x \geq 1 - t/2 - 4\varepsilon > 1/2$. The F/G modifications are therefore absent at every newly chosen lottery report. Bidder 1 can only take the scarce singleton, at price $P \geq \max(x, x + y - c)$. If $t > P$, the displayed lottery inequalities force $(x, y) \in S$ and

$$t - P \leq t - \max(x, x + y - c) < \varepsilon.$$

The entry fee selects empty, releasing that item. Its low item is already free because its value is at most $1/5 < c$. Equalities use the explicit empty priority. The earlier splices have the full same-orientation, opposite-orientation and reverse-Q capacity checks in the certificate proof notes; these checks include the excluded $\rho = c$ and $\sum v = b$ faces. Finite menus with Borel parameters yield joint measurability and bounded payments. Feasible marginals can be realized itemwise. $\square$

This release preserves the full F and Q certificates. It changes some Eplus menus and their binding residuals. A sufficient preserved Eplus scope excludes S for bidder 1 and excludes $\max(w) \in [.7 - 4\varepsilon, .71 + \varepsilon]$ for bidder 2. The wider second exclusion includes releases outside W. No conditional optimality is asserted on excluded fibers, and no common auction support is obtained.

6.5 Exact new revenue and a sub-0.01 bracket

Let $R_{4.5}$ be the exact revenue of the source-bound reference, specified by the included V4.02 continuous integrals plus 472617707798758460108653/101062797120000000000000000000. The

Theorem 10 (Stream weak duality). *For each bidder i , let G_i be a jointly C^1 vector field on $[0, 1]^4$. Suppose that, for each opponent report, its own-type divergence is zero and its normal component vanishes on every side of the own-type square. Put $F_i(v) = F^0(v_i) + G_i(v)$. Every admissible mechanism M satisfies*

$$\text{Rev}(M) \leq \mathcal{E}(F) := \int_{[0,1]^4} \sum_{j=1}^2 \max\{0, F_{1j}(v), F_{2j}(v)\} dv. \quad (8)$$

Consequently $\text{OPT} \leq \mathcal{E}(F)$.

Proof. Fix bidder i and an opponent report on a full-measure set for which the conditional expected-payment slice is integrable. Write truthful utility as $u(v)$, expected allocation as $X(v) \in [0, 1]^2$, and expected payment as $p(v)$. Pointwise DSIC gives

$$u(v) \geq u(r) + (v - r) \cdot X(r) \quad \text{for every } v, r.$$

Thus u is convex and $X(r) \in \partial u(r)$. These inequalities and $0 \leq X_j \leq 1$ also make u Lipschitz. At every differentiability point, $X = \nabla u$. At nonsmooth points we retain the mechanism's given jointly measurable subgradient selection; its pointwise feasibility is part of the model, not inferred from an almost-everywhere gradient.

Joint measurability and the assumed integrability of the payment, together with boundedness of $v \cdot X$, make $u = v \cdot X - p$ measurable and integrable. Thus the conditional identities below apply on this full-measure set of opponent reports and may then be integrated by Fubini.

We retain the origin utility explicitly. The Lipschitz bound gives $|u(v) - u(0)| \leq 2$, so the opponent-dependent value $u(0)$ is integrable as well. Ex-post IR gives $u(0) \geq 0$.

Partition the square into the two radial charts

$$v = s(1, t) \quad \text{and} \quad v = s(t, 1), \quad 0 \leq s, t \leq 1,$$

each with Jacobian s . On a fixed ray set $q = (1, t)$ or $(t, 1)$ and $g(s) = u(sq)$. Convexity makes g absolutely continuous and $g'(s) = q \cdot X(sq)$ almost everywhere and $g(0) = u(0)$. Since $p(sq) = sg'(s) - g(s)$,

$$\int_0^1 p(sq) s ds = \int_0^1 \frac{3s^2 - 1}{2} g'(s) ds - \frac{u(0)}{2}.$$

Combining the two charts gives the conditional revenue identity

$$\int_{[0,1]^2} p(v) dv = \int_{[0,1]^2} F^0(v) \cdot X(v) dv - u(0).$$

The correction is bounded, and $u \in W^{1,\infty}$. Weak Green integration, zero own-type divergence, and the zero normal trace therefore give

$$\int_{[0,1]^2} G_i \cdot \nabla u = 0.$$

Thus replacing F^0 by F_i preserves the conditional identity.

Apply this identity to both bidders and then integrate over opponent reports. Fubini is justified by the integrability established above. Dropping the nonpositive origin terms leaves the sum of the field-allocation pairings. For every profile and item j , feasibility gives $X_{1j} + X_{2j} \leq 1$, with both allocations nonnegative. Hence

$$\sum_i F_{ij} X_{ij} \leq \max\{0, F_{1j}, F_{2j}\}.$$

This pointwise relaxation is valid for arbitrary expected allocations and therefore for arbitrary internal randomization. Integrating proves (8); the maximizing allocation in its integrand need not itself be implementable by a DSIC mechanism. $\square$

Boundary interpretation. The radial identity can also be checked by integration by parts. The classical divergence of F^0 is 3 in each open triangle $x > y > 0$ and $y > x > 0$, hence almost everywhere. Continuity across the diagonal away from the origin makes the interface fluxes cancel. The outer boundary flux is 2, while the flux around a puncture at the corner $(0,0)$ tends to 1, with the normal pointing into the removed square. Applying Green's formula to the two triangles outside $[0, \varepsilon]^2$ and letting $\varepsilon \downarrow 0$ gives

$$\int F^0 \cdot \nabla u = \int_{x=1} u + \int_{y=1} u - 3 \int_{[0,1]^2} u + u(0).$$

Lipschitz continuity replaces the trace at the puncture by $u(0)$ with vanishing error. This accounts for the corner term without treating a boundary corner as an interior distributional source.

7.2 An uncorrected radial benchmark

Taking $G_i = 0$ gives a useful analytic benchmark. In the radial charts, $F^0(sq) = \phi(s)q$, where $\phi(s) = (3s^2 - 1)/(2s)$.

For completeness, the remaining scalar expectation can be evaluated exactly. Under the two radial charts, a uniform type can be generated by taking S with density $2s$, an independent fair choice of which coordinate is maximal, and an independent $T \sim U[0, 1]$. For one fixed item its radial coordinate therefore has distribution

$$Q \sim \frac{1}{2}\delta_1 + \frac{1}{2}U[0, 1].$$

Set $Y = \max\{0, \phi(S)\}$, independently of Q . For two bidders let $a = \max(Y_1, Y_2)$ and $b = \min(Y_1, Y_2)$. Conditioning on the four atom/uniform cases for Q_1, Q_2 gives

$$\mathbb{E}_{Q_1, Q_2} \max\{Y_1 Q_1, Y_2 Q_2\} = \frac{3}{4}a + \frac{b^2}{6a},$$

where b^2/a is defined as zero when $a = 0$. Since the two items are symmetric,

$$U_{\text{rad}} = \frac{3}{2}\mathbb{E}[a] + \frac{1}{3}\mathbb{E}\left[\frac{b^2}{a}\right]. \quad (9)$$

Put $r = 1/\sqrt{3}$. The function ϕ is increasing from zero to one on $[r, 1]$, while $Y = 0$ below r . The larger of two independent S 's has density $4s^3$, hence

$$\mathbb{E}[a] = \int_r^1 4s^3 \phi(s) ds = \frac{8}{15} + \frac{4}{45\sqrt{3}}.$$

Ordering the two S 's gives the second moment as the explicit integral

$$\mathbb{E}\left[\frac{b^2}{a}\right] = 8 \int_r^1 \frac{s}{\phi(s)} \left(\int_r^s t \phi(t)^2 dt \right) ds.$$

The inner integral is

$$J(s) = \frac{9}{16}s^4 - \frac{3}{4}s^2 + \frac{1}{4}\log s + \frac{3}{16} + \frac{1}{8}\log 3,$$

so the last display equals

$$16 \int_r^1 \frac{s^2 J(s)}{3s^2 - 1} ds.$$

Set $x = \sqrt{3}s$. The cancellation of the $\log 3$ terms in J reduces this integral to

$$\frac{1}{3\sqrt{3}} \int_1^{\sqrt{3}} x^2(x^2 - 3) dx + \frac{4}{3\sqrt{3}} \int_1^{\sqrt{3}} \frac{x^2 \log x}{x^2 - 1} dx.$$

Use $x^2/(x^2 - 1) = 1 + 1/(x^2 - 1)$. For the only non-elementary term, the substitution $y = (x - 1)/(x + 1)$ has upper endpoint $d = 2 - \sqrt{3}$ and gives

$$\int_1^{\sqrt{3}} \frac{\log x}{x^2 - 1} dx = \frac{1}{2} \int_0^d \frac{\log(1 + y) - \log(1 - y)}{y} dy = \frac{1}{2} [\text{Li}_2(d) - \text{Li}_2(-d)],$$

where

$$\frac{d}{dz} \text{Li}_2(z) = -\frac{\log(1 - z)}{z}$$

was used in the last equality. The remaining polynomial and $\int \log x dx$ terms then give

$$-\frac{26}{15} + \frac{8}{5\sqrt{3}} + \frac{4}{3} \log \sqrt{3} + \frac{2}{3\sqrt{3}} [\text{Li}_2(d) - \text{Li}_2(-d)].$$

Substitution in (9) gives

$$U_{\text{rad}} = \frac{2}{9} + \frac{2}{3\sqrt{3}} + \frac{4}{9} \log \sqrt{3} + \frac{2}{9\sqrt{3}} [\text{Li}_2(2 - \sqrt{3}) - \text{Li}_2(\sqrt{3} - 2)].$$

This gives $\text{OPT} \leq U_{\text{rad}} = 0.920577474278749\dots$. We next improve this benchmark with a rational stream correction.

7.3 The rational stream witness

We now add an opponent-dependent curl. For an exponent $e = (e_0, e_1, e_2, e_3)$, put

$$m_e = x^{e_0} y^{e_1} z^{e_2} w^{e_3}, \quad \sigma(e) = (e_1, e_0, e_3, e_2).$$

Let E be the lexicographically smaller member of every nontrivial σ -pair of total degree at most four. Define

$$h(x, y, z, w) = \sum_{e \in E} \theta_e (m_e - m_{\sigma(e)}), \quad \psi = x(1 - x)y(1 - y)h, \quad G = (\partial_y \psi, -\partial_x \psi).$$

The thirty-two rational coefficients are listed below in the exact basis order used by the verifier. A convex quasi-Monte Carlo search was used only to discover them; the proof defines the field by the following rational numbers and does not trust the numerical optimization.

e	θ_e	e	θ_e
(0, 0, 0, 1)	-2995385/993184	(0, 0, 0, 2)	2337131/676164
(0, 0, 0, 3)	5724621/964727	(0, 0, 0, 4)	-4591777/714341
(0, 0, 1, 2)	-834567/846478	(0, 0, 1, 3)	6291553/970203
(0, 1, 0, 0)	2762675/446201	(0, 1, 0, 1)	206964/961681
(0, 1, 0, 2)	-6090207/979156	(0, 1, 0, 3)	6862090/913753
(0, 1, 1, 0)	-5206994/854575	(0, 1, 1, 1)	515889/117356
(0, 1, 1, 2)	-3346363/843816	(0, 1, 2, 0)	-125263/465506
(0, 1, 2, 1)	3005651/895053	(0, 1, 3, 0)	-1043238/468269
(0, 2, 0, 0)	1660919/994155	(0, 2, 0, 1)	1276309/943222
(0, 2, 0, 2)	-2075136/255347	(0, 2, 1, 0)	404816/59269
(0, 2, 1, 1)	250988/352269	(0, 2, 2, 0)	3825425/994154
(0, 3, 0, 0)	-5383501/354845	(0, 3, 0, 1)	3312081/940771
(0, 3, 1, 0)	-4885471/759831	(0, 4, 0, 0)	7648954/974695
(1, 1, 0, 1)	-603823/508896	(1, 1, 0, 2)	417332/188831
(1, 2, 0, 0)	17884893/955196	(1, 2, 0, 1)	992961/785407
(1, 2, 1, 0)	2761171/712822	(1, 3, 0, 0)	-3841999/611370

Give bidder 2 the same construction with (x, y) and (z, w) swapped, and put $F_i = F_i^0 + G_i$.

Lemma 11 (Rational witness feasibility). *These fields satisfy Theorem 10. They are equivariant under bidder exchange and simultaneous item exchange.*

Proof. The corrections are polynomials. Equality of mixed partial derivatives gives $\text{div}_{x,y} G = 0$. Moreover $G \cdot n = 0$ on each side of the own-type square: on a vertical side $\partial_y \psi = 0$, and on a horizontal side $-\partial_x \psi = 0$, because of the boundary factors in ψ . Bidder symmetry follows from the definition of G_2 . Finally, $h(y, x, w, z) = -h(x, y, z, w)$, hence the same antisymmetry holds for ψ . Differentiation gives

$$G(y, x, w, z) = (G_y(x, y, z, w), G_x(x, y, z, w)),$$

where G_x, G_y denote the two components. The radial field has the same item-exchange property, proving the claim. $\square$

7.4 Finite exact enclosure

Proposition 12 (Certified envelope). *For the fields in Lemma 11, $\mathcal{E}(F) \leq U_{\text{stream}}$.*

Computer-assisted proof. It remains to bound the integral in (8) exactly. Split each bidder square into (s, st) and (st, s) . In each of the four chart pairs, multiplication by the Jacobian $s_1 s_2$ turns both item-1 competitors into exact rational polynomials in (s_1, t_1, s_2, t_2) . The anti-symmetry of h under $(x, y, z, w) \mapsto (y, x, w, z)$ exchanges the two field components; invariance of the uniform distribution therefore makes the item-2 integral identical. The two charts overlap only on the diagonal and at the origin, which are null sets.

The certificate converts each polynomial and their exact difference to tensor Bernstein form. A polynomial lies in the convex hull of its Bernstein control coefficients, and its box integral is the box volume times their mean. Every exact initial control c is stored as an integer $m = \lfloor 10^9 c \rfloor$ with error radius one. A degree- n de Casteljau half-box split makes n floor averages, so the rigorous radius update is $E \leftarrow E + n$. Lower bounds for a competitor and for the exact competitor difference certify a fixed winner; otherwise the maximum upper control gives a valid supremum charge.

The tree uses the maximum-variation axis above the final four levels of a depth-21 base partition and exact one-step child-bound lookahead in those final four levels. Each unresolved depth-21 leaf is split once along the best exact child-bound axis if and only if that split saves at least one integer accumulator unit; otherwise its depth-21 charge is retained. Each unresolved child of an accepted depth-22 split is then tested independently: its best depth-23 split is accepted if and only if it saves at least one further unit, and otherwise its depth-22 charge is retained. Taking the minimum of several valid upper bounds is valid. All contributions are accumulated over the common depth-23 dyadic denominator. Each accepted split replaces its parent charge by charges on two children with disjoint interiors; a retained box is charged once. The coverage check sums the leaf volumes over all four chart pairs to $4 \cdot 2^{23} = 33554432$ common-denominator units.

The manifest-driven verifier reconstructs the polynomials and reports

boxes fixed before unresolved depth-21 leaves : 404804,
 unresolved depth-21 leaves : 462796,
 leaves refined to depth 22 : 391618,
 depth-21 leaves retained : 71178,
 fixed depth-22 children : 173198,
 unresolved depth-22 children : 610038,
 depth-22 children refined : 609951,
 depth-22 children retained : 87,
 fixed depth-23 children : 249438,
 unresolved depth-23 children : 970464,
 integer accumulator : 3715139591287203.

After the four chart pairs and the item-symmetry factor two, this gives

$$\int \sum_j \max\{0, F_{1j}, F_{2j}\} \leq \frac{2 \cdot 3715139591287203}{10^9 2^{23}} = \frac{3715139591287203}{4194304000000000}.$$

□

Together with Theorem 10 and Lemma 11, this proves the upper half of Theorem 1. The verifier and its clean transcript are in the following certificate directory:

`certificate/continuous_stream_degree4_two_level_nonuniform_upper_bound/`

7.5 Reproducibility and trusted computations

The publication archive separates the proof inputs from the numerical search that found them. The trusted upper-bound directory contains the rational coefficient manifest, polynomial constructor, exact verifier, non-importing replay, and captured transcripts; the root SHA-256 manifest covers these files. The verifier checks all manifest expectations and fails closed on any mismatch. Before fixed-point conversion, every polynomial and Bernstein coefficient is constructed with `fractions.Fraction`. The largest axis degree is eight, the maximum initial absolute control is 1475346733, and the maximum observed directed-rounding radius is 181 fixed-point units. Signed array operations are checked against $2^{63} - 1$; child charges and the global accumulator use arbitrary-precision Python integers.

The non-importing iterative replay independently reconstructs the basis, chart maps, polynomial arithmetic, Bernstein conversion, subdivision, winner tests, selective-refinement rule, and coverage accounting. Both implementations visit 3,738,334 retained-tree nodes and reproduce all refinement counts, with

$$\text{accumulator} = 3715139591287203, \quad \text{coverage} = 33554432/33554432.$$

The independence is at the implementation level: both programs share the rational input manifest and the Bernstein enclosure principle. They audit the finite arithmetic; Theorem 10 supplies the analytic weak-duality argument.

The lower-bound chain reconstructs the 144 affine/pivot cells with one exact base verifier. Separate replays recompute the added gains from twenty common-fee rows, 41 positive bundle-pivot cells, and eight item-containment rows, while retaining the shared base revenue. They use scalar rational arithmetic, finite-difference consistency checks, exact Boole quadrature, and, for the final non-common perturbation, exact demand-polygon clipping. The degree bounds needed for exact quadrature follow from the analytic menu formula and the verified price regions, not from finite sampling. The new source-bound lower package separately replays the 29 declared

mathematical entry points and independent audits of the final menu, coordinate-line supports, and exact radical/logarithmic increments. It retains the earlier certified reference integral explicitly. This is not a claim of a second from-scratch integration of every historical base layer. Its supported replay runs from the exported package without access to the research workspace. The supported release entry point is `python -B verification/reproduce_all.py`. It reruns all publication-facing checks, compares their structured outputs with the theorem endpoints, and verifies the archive hash manifest. The computational trusted base comprises Python’s integer semantics, `fractions.Fraction`, the checked NumPy integer operations, JSON parsing, SHA-256, and file I/O. The entry point rejects optimized Python execution with an explicit runtime check and a nonzero exit status; this protects the executable assertions retained in predecessor kernels. Direct kernel invocations also require non-optimized Python. No optimizer output is read on the trusted proof path.

8 Discussion and open problem

The new lower construction brings the certified gap to 0.0099384483702321295386 or less, below 0.01, while leaving the global upper witness unchanged. Relative to the retained deterministic revenue 0.8746049864028362682920..., the increase is 0.001214867745586187.... This numerical improvement is accompanied by a structural reduction: full randomized conditional optimization is solved on the stated residual regions, and the corresponding capacity measures identify admissible ownership changes. The reported GemNet revenue of approximately 0.876 remains higher. Our result concerns an explicit mechanism, exact accounting and proof-supported primal-dual structure.

The diagonal and lottery supports show how information-rent constraints can place capacity prices on report lines as well as on positive-volume allocation regions. A finite menu emerges as an equality case of an unrestricted conditional bound. The resulting support is not automatically common to both bidders. Indeed, the complete corner release improves a reference whose conditional lottery menu was already optimal against its old residual. The revenue effect must be integrated over every report forced to change by incentives, including buyers who retain allocations but pay different prices.

These theorems do not classify optimal auctions, prove a finite allocation range sufficient in general, or establish optimality under unrestricted outer reallocations. The final mechanism preserves the F and Q conditional certificates while changing equality conditions on part of Eplus. A common capacity measure and compatible incentive system supporting both bidders remain missing. The older stream envelope also relaxes convex-gradient compatibility, so a pointwise maximizing allocation need not be DSIC. Closing the remaining gap requires these allocation, rent and capacity conditions to agree. Neither higher polynomial degree alone nor stationarity inside the displayed finite family supplies that conclusion.

Data, code, and materials availability

The project repository is publicly accessible at

<https://github.com/Baymax-ray/two-bidder-two-item-dsic-certificates>.

The accompanying release archive contains all exact certificate data, deterministic verification code, transcripts, environment information, and SHA-256 manifests needed to reproduce the stated bounds. Manuscript, certificate data, and other textual materials are released under CC BY 4.0. Verification software is separately released under the MIT License. Discovery-only experiments are segregated from the trusted proof artifacts.

AI-assisted tools declaration

This manuscript was developed with the assistance of OpenAI GPT-5.6 Sol and OpenAI Codex. The tool assisted with mathematical exploration, numerical candidate generation, implementation and debugging of certificate and verification code, and drafting and revision of the manuscript. Separately scoped AI-assisted audits checked the new proofs, source implementations and exact arithmetic; these are internal checks, not external peer review. AI-generated suggestions and numerical search outputs were not treated as proofs; the stated results are supported by the analytic arguments and exact certificates described here. The author retains responsibility for the content, correctness, and final presentation of this work.

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